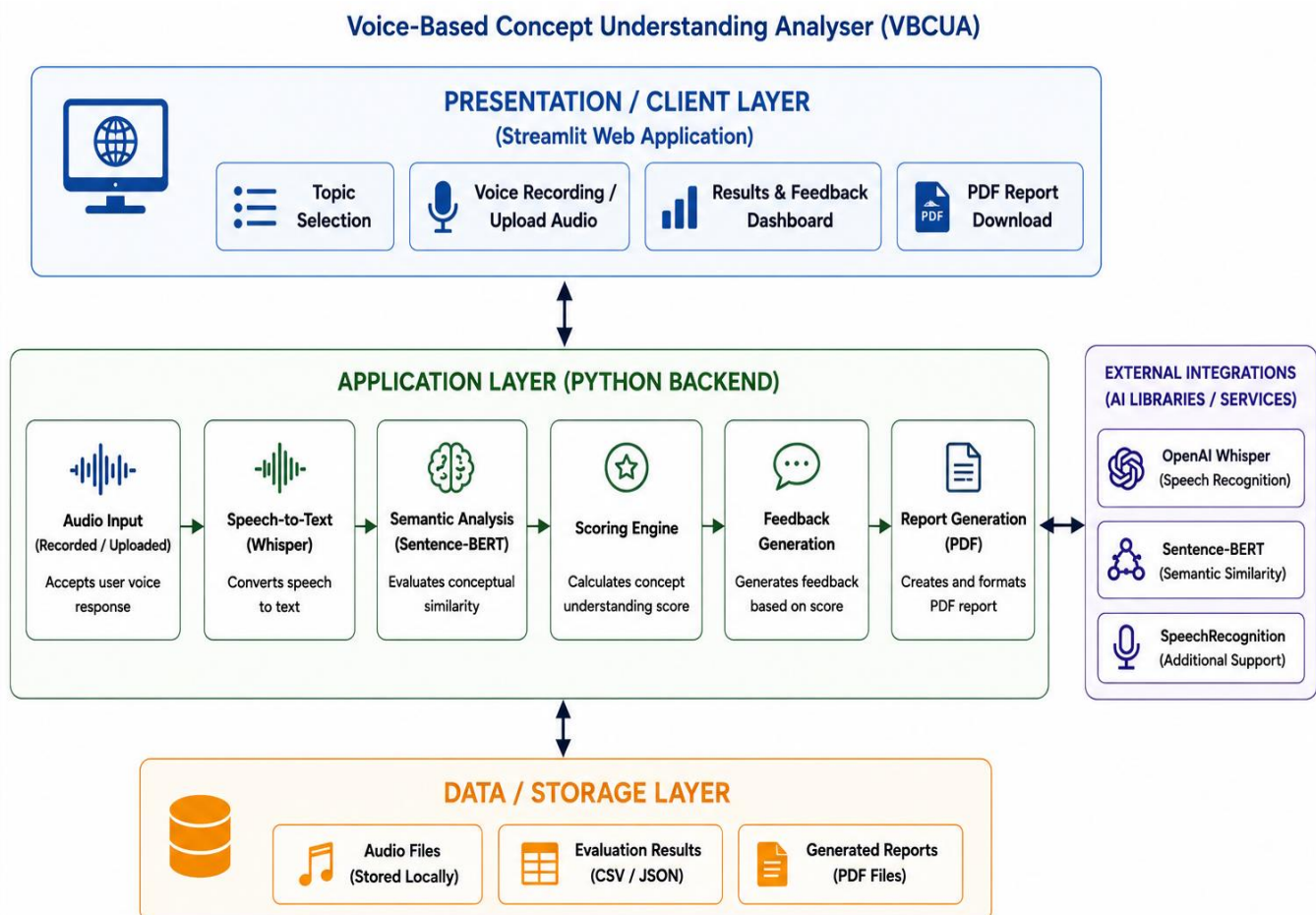


Solution Architecture

Date	07 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Solution Architecture is the practice of designing, describing, and managing the engineering configuration of a software system to solve a specific business problem. It acts as the bridge between business requirements and technical implementation.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Provides the graphical user interface for recording/uploading speech, selecting topics, displaying scores, feedback, and downloadable reports.	Streamlit
<i>Application Layer</i>	Processes speech, converts audio to text, performs semantic similarity analysis, calculates concept understanding scores, generates feedback, and creates PDF reports.	Python, OpenAI Whisper, Sentence-BERT, Librosa, PyDub, Pandas, NumPy, ReportLab
<i>Data Storage Layer</i>	Stores uploaded audio files, evaluation results, generated reports, and temporary application data.	Local File Storage, CSV, JSON, PDF
<i>External AI Libraries</i>	Supplies speech recognition and natural language processing capabilities required for evaluating spoken responses.	OpenAI Whisper, Sentence-BERT, SpeechRecognition